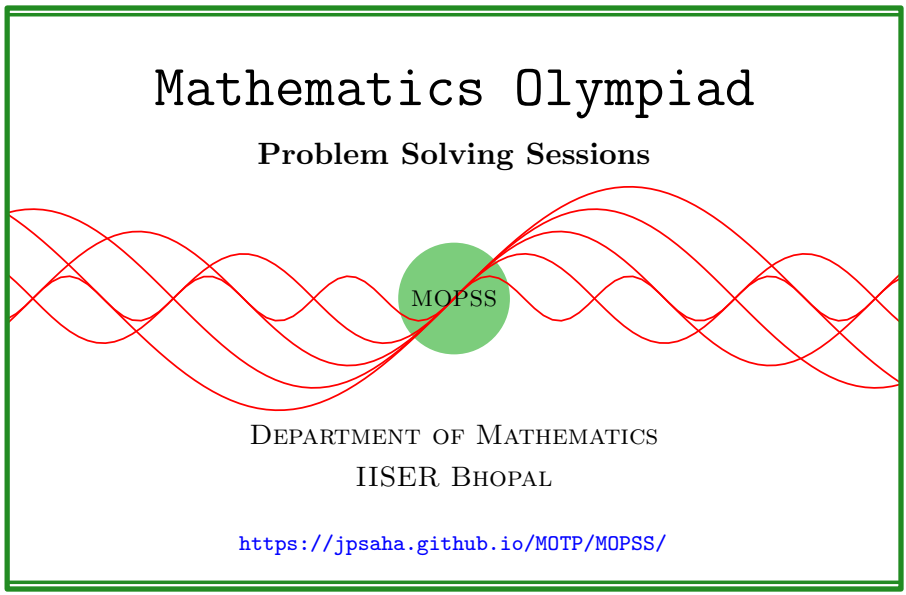
Differentiation and multiple roots

MOPSS



Suggested readings

- Evan Chen's advice On reading solutions, available at https://blog.evanchen.cc/2017/03/06/on-reading-solutions/.
- Evan Chen's Advice for writing proofs/Remarks on English, available at https://web.evanchen.cc/handouts/english/english.pdf.
- Notes on proofs by Evan Chen from OTIS Excerpts [**Che25**, Chapter 1].
- Tips for writing up solutions by Edward Barbeau, available at https://www.math.utoronto.ca/barbeau/writingup.pdf.
- Evan Chen discusses why math olympiads are a valuable experience for high schoolers in the post on Lessons from math olympiads, available at https://blog.evanchen.cc/2018/01/05/lessons-from-math-olympiads/.

List of problems and examples

1 Exercise (Putnam 1956 B7, cf. IMOSL 1981 Cuba) 2
2 Exercise (Putnam 2018 B2,) 4

§1 Differentiation and multiple roots

Lemma 1

Let $P(x)$ be a polynomial with complex coefficients, and α be a complex number. If α is a root of $P(x)$ of multiplicity $k \geq 1$, then α is a root of $P'(x)$ of multiplicity $k-1$. Moreover, α is a root of $P(x)$ having multiplicity at least $r \geq 2$ (i.e., $(x-\alpha)^r$ divides $P(x)$) if and only if it is a root of $P(x), P'(x), \ldots, P^{(r-1)}(x)$, where $P^{(k)}(x)$ denotes the k-fold derivative of $P(x)$.

Solution 1. ■

Exercise 1 (Putnam 1956 B7, cf. IMOSL 1981 Cuba)**.** The polynomials $P(z)$ and $Q(z)$ with complex coefficients have the same set of numbers for their zeroes but possibly different multiplicities. The same is true of the polynomials $P(z)+1$ and $Q(z)+1$. **Assume that at least one of $P(z), Q(z)$ is nonconstant.** Prove that $P(z) = Q(z)$.

Walkthrough —

(a) Assume that $\deg P \geq \deg Q$.

(b) Denote these two set of roots by S_1, S_2. Considering multiplicities, show that

$$2\deg P - |S_1| - |S_2| \leq \deg P' = \deg P - 1,$$

which yields

$$|S_1| + |S_2| > \deg P.$$

(c) Note that $P-Q$ vanishes at the elements of $S_1 \cup S_2$, which has size larger than the degree of $P-Q$.

Solution 2. On the contrary, let us assume that P, Q are not equal. Without loss of generality, let us assume that $\deg P \geq \deg Q$. Let n denote the degree of P. Let us denote by j the number of distinct roots of P, and by k the number of distinct roots of $P+1$. Note that these two sets of roots are disjoint. Using the equality

$$P(z) - Q(z) = (P(z)+1) - (Q(z)+1),$$

it follows that $P-Q$ has at least $j+k$ distinct roots. Since $\deg P \geq \deg Q$, it follows that $\deg(P-Q) \leq \deg P = n$, and hence $j+k \leq n$.

The content posted here and at this blog by Evan Chen are quite useful.

Note that the derivative P' has at least $n-j$ roots common with P counted with multiplicity, and at least $n-k$ roots common with $P+1$ counted with multiplicity. Since these two sets of roots are disjoint, it follows that

$$\deg P' \geq 2n - j - k \geq n.$$

Since $\deg P \geq \deg Q$ and one of P, Q is nonconstant, it follows that P is nonconstant, and hence $\deg P' = n-1$. Thus, we have

$$n - 1 = \deg P' \geq n,$$

which is a contradiction. Thus, we obtain $P = Q$. ■

Theorem 2 (Gauss–Lucas)

Let P be a nonconstant polynomial with complex coefficients. Then, all the roots of P' lie in the convex hull of the set of roots of P.

Proof. Write P as

$$P(z) = a(z - z_1)(z - z_2)\cdots(z - z_n),$$

where a is a nonzero complex number, and $z_1, z_2, \ldots, z_n$ are the roots of P counted with multiplicity. Note that

$$P'(z) = a\sum_{k=1}^{n}(z - z_1)\cdots(z - z_{k-1})(z - z_{k+1})\cdots(z - z_n).$$

Thus, for any z which is a root of P' but not of P, we have

$$\begin{aligned}\frac{P'(z)}{P(z)} &= \sum_{k=1}^{n}\frac{1}{z - z_k} \\ &= \sum_{k=1}^{n}\frac{\overline{z - z_k}}{|z - z_k|^2},\end{aligned}$$

which yields

$$\sum_{k=1}^{n}\frac{z}{|z - z_k|^2} = \sum_{k=1}^{n}\frac{z_k}{|z - z_k|^2}.$$

This shows that if z is a root of P' but not of P, then

$$z = \sum_{k=1}^{n}\frac{\dfrac{1}{|z - z_k|^2}}{\sum_{\ell=1}^{n}\dfrac{1}{|z - z_\ell|^2}}z_k,$$

which is a convex combination of the roots of P. Further, if z is a root of both P' and P, then z lies in the convex hull of the roots of P. This completes the proof. □

Exercise 2 (Putnam 2018 B2,). Let n be a positive integer, and let $f_n(z) = n + (n-1)z + (n-2)z^2 + \cdots + 2z^{n-2} + z^{n-1}$. Show that all the roots of $f_n(z)$ lie outside the closed unit disk $\{z \in \mathbb{C} : |z| \leq 1\}$.

> **Walkthrough** —
> **(a)**

Solution 3. Let us assume that there exists a root z of f_n such that $|z| \leq 1$. Note that

$$(1-z)f_n(z) = n - z - z^2 - \cdots - z^{n-1} - z^n.$$

Using the triangle inequality, we have

$$\begin{aligned} 0 = |(1-z)f_n(z)| &= |n - z - z^2 - \cdots - z^{n-1} - z^n| \\ &\geq n - |z + z^2 + \cdots + z^{n-1} + z^n| \\ &\geq n - |z| - |z^2| - \cdots - |z^{n-1}| - |z^n| \\ &> n - 1 - 1 - \cdots - 1 - 1 \\ &= 0. \end{aligned}$$

This shows that $z, z^2, \ldots, z^n$ have absolute value 1, and have the same argument, and hence $z = 1$. However, $f_n(1) = n + (n-1) + (n-2) + \cdots + 2 + 1$, which is nonzero. This is a contradiction. Hence, all the roots of f_n lie outside the closed unit disk. ■

Solution 4. Let $g_n(z)$ denote the polynomial $z^n + z^{n-1} + \cdots + z + 1$. Note that

$$g_n'(z) = nz^{n-1} + (n-1)z^{n-2} + \cdots + 2z + 1 = z^{n-1} f_n\left(\frac{1}{z}\right).$$

Observe that the roots of the polynomials g_n', f_n are nonzero, and a nonzero complex number z is a root of f_n if and only if $\frac{1}{z}$ is a root of g_n'. By the Gauss–Lucas theorem (Theorem 2), all the roots of g_n' lie in the convex hull of the set of roots of g_n. Note that the roots of g_n are precisely the n-th roots of unity other than 1. Since g_n has no multiple roots, it follows that g_n' does not vanish at any of the roots of g_n. Thus, all the roots of g_n' lie in the interior of the convex hull of the set of roots of g_n, and consequently, the roots of g_n' lie in the open unit disk. Therefore, all the roots of f_n lie outside the closed unit disk. ■

References

[**Che25**] Evan Chen. *The OTIS Excerpts*. Available at https://web.evanchen.cc/excerpts.html. 2025, pp. vi+289 (cited p. 1)